KIFARU PORTOS

Kenya Intelligent Freight, Access, Routing and Unified Logistics Platform

Project Proposal to the Judging Panel

Rolling Clearance System v2: port-anchored clearance for heavy trucks on the Mombasa - Northern Corridor route, with implementation and simulation plan

Item	Detail
Version	1.0 (proposal), 05 November 2026
Prepared by	KIFARU PORTOS project team
Prepared for	The judging panel; and, for implementation, KPA, KRA, KeNHA, NTSA, KenTrade, KEBS, the weights-and-measures body, the Kenya Transporters Association and candidate loaders
Live platform	https://kifaru-portos-1-0.vercel.app
Source repository	https://github.com/Jrjunior-mk/KIFARU-PORTOS-1.0
Status of figures	Two kinds of numbers, always labelled: published figures (KPA, KeNHA, NCTTCA, SCEA, KenTrade) are cited with period and source; stop shares, truck-hours, dwell and queue results are modelled from stated assumptions and are not measurements. Phase 0-2 replaces modelled figures with observed data before any benefit is claimed.

Contents

Contents 2

1. Executive summary 4

1.1 At a glance for the judges . . . 4

2. Background and problem statement 5

2.1 The corridor today . . . 5

2.2 Where the chain duplicates checks and leaves gaps . . . 5

2.3 Problem statement . . . 5

2.4 The evidence: what published data shows . . . 6

3. Objectives 6

4. Proposed solution: RCS v2 7

4.1 Three truck streams, one rule . . . 7

4.2 The Load Compliance Token . . . 7

4.3 Parameter coverage beyond weight . . . 8

4.4 The Rolling Verification Lane . . . 8

4.5 Decision rules . . . 8

4.6 What changes from the earlier proposal . . . 9

4.7 SmartPort: the eleven-stage port visit . . . 9

5. Legal and regulatory basis 9

6. System architecture and current implementation 10

6.1 Architecture . . . 10

6.2 What is already built . . . 11

6.3 Data provenance . . . 11

7. Simulation methodology 11

7.1 Questions the simulation answers . . . 11

7.2 Four linked levels . . . 12

7.3 The two L2 models used in this proposal . . . 12

7.4 Experiment design, verification and validation . . . 12

8. Simulation results (modelled) 14

8.1 Port trucks: effect by average loading . . . 14

8.2 Mixed traffic: effect by token coverage . . . 14

8.3 Sensitivity . . . 15

8.4 Replication and confidence intervals . . . 16

8.5 Go/no-go gate . . . 16
8.6 Station queue model (SimPy) . . . 17
8.7 Port process: existing operations vs SmartPort . . . 18
8.8 Not yet counted . . . 19

9. Implementation roadmap **20**
9.1 Simulation work plan (16 weeks, Phase 2) . . . 20
9.2 Integration map . . . 20
9.3 Deliverables by phase . . . 20
9.4 Cost drivers . . . 21

10. Risks and mitigations **21**

11. Monitoring and evaluation **21**

12. Decisions requested **22**

Appendix A. Data requests **23**

Appendix B. Open questions to confirm **23**

Appendix C. Model assumptions **23**

Appendix D. Sources **23**

1. Executive summary

Heavy trucks leaving the Port of Mombasa, and trucks loaded at mines, quarries, factories and depots, are checked many times for identity, cargo and release status, yet their weight is first measured on the highway, where an overload is hardest and most expensive to fix, and the same truck is then weighed again at every weighbridge on its route. KIFARU PORTOS proposes RCS v2, port-anchored clearance: weigh every heavy truck once at its first trusted point, issue a signed Load Compliance Token, and let every later station honour that token. For port cargo the first trusted point is the port exit; for industrial cargo it is an approved loading-site scale; for everyone else it is the first road station. Doubtful trucks roll through a Rolling Verification Lane at about 10 km/h where weight, dimensions, lifted axles and per-wheel loads are captured without a stop. Red cases still go to the static scale and an authorised officer, as the EAC Vehicle Load Control (Enforcement Measures) Regulations, 2018 require.

The platform that implements this design is already built in software and running: a FastAPI backend with a deterministic clearance engine, a clearance registry (tokens, release orders, permits, demerits), a React web application with role dashboards, a public verification page and a 3D digital twin, a roadside gateway, firmware and hardware designs, and a simulation package. It is deployed at https://kifaru-portos-1-0.vercel.app on a hosted PostgreSQL database, with 177 automated tests passing.

Modelled effect (to be confirmed with field data):

- At 85 % average loading, static-scale visits by port trucks at the first road station fall from 23.2 % of trucks under road-only screening to 5.3 % with RCS v2, while 14.8 % are re-weighed inside the port where cranes and customs are at hand. That is about 143 truck-hours a day of road queueing removed at 8 000 trucks a day.
- For the full traffic mix, static-scale visits at the first station fall from 20.8 % to 10.0 %, 7.1 %, 4.3 % and 1.5 % as token coverage rises from 30 % to 90 %; later stations send almost nothing to a static scale (0.04 %).
- One rolling verification lane handles about 370 trucks an hour, roughly 25 static lanes; the modelled peak demand on it is 116 trucks an hour.
- No truck over the limit is modelled as cleared without a stop (false-clear rate 0.000 %); real fraud such as plate swaps is handled by audits and token binding, outside the model.
- The go/no-go gate set for the simulation phase (road stop share cut by at least 10 points, false clear at most 1 %, port re-weigh volume practical) is met in the model; the gate is re-evaluated on measured data before any hardware spend.

SmartPort, the port-side layer, follows every truck through eleven stages from pre-arrival registration to delivery, with the data each stage needs, the agency that owns it, and dwell-time KPIs set against published benchmarks. Calibrated to KPA's published 2023 truck turnaround of 6.7 hours, the port model shows mean port dwell falling from 6.8 h to 4.0 h (-42 %) and weighbridge service from 5.1 to 1.6 minutes (modelled).

1.1 At a glance for the judges

Criterion	What this proposal shows	Where
Problem significance	The Port of Mombasa handled 45.45 million tonnes in 2025; Mariakani weighed 7,954-9,184 trucks a day in Jul-Sep 2026; import container dwell was 87 h against a 60 h target and Mombasa-Nairobi transit 66 h against 29 h (published, NCTTCA).	Section 2.4
Innovation	Weigh once at the first trusted point and honour a signed Load Compliance Token downstream; a rolling verification lane replacing about 25 static lanes; one registry shared by all agencies; every number tagged with its provenance.	Sections 4, 6.3
Feasibility	Not a concept: the platform is built, tested (177 automated tests) and live, on a hosted database; legal hooks identified regulation by regulation.	Sections 5, 6
Impact (modelled)	Road static-scale stops 23.2 % to 5.3 % for port trucks; first-station static visits 20.8 % to 1.5 % at 90 % token coverage; port dwell -42 %.	Section 8
Evidence and rigour	52-entry data register from KPA, KeNHA, NCTTCA, SCEA and KenTrade; Monte Carlo and discrete-event models with replications, sensitivity and a go/no-go gate.	Sections 2.4, 7, 8

Criterion	What this proposal shows	Where
Scalability	Same token and registry serve every weighbridge on the Northern Corridor and inland depots; cloud platform; agency adapters for KPA, KRA, KenTrade, KEBS, KeNHA, NTSA.	Sections 6, 9
Sustainability	Partner model allowed by Reg 3(3); loaders and fleets gain by avoiding stops, fees and detention; staged spend gated on validated results.	Sections 5, 9.4
Risks	Named with mitigations: port access, scale approval, origin under-reporting, token fraud, land for lanes, model risk.	Section 10

Decisions requested: approve the redesign around the port gate-out node and the token; approve Phases 0-2 including the 16-week simulation and baseline study with a gate before hardware spend; authorise approaches to KPA, KRA, KeNHA, NTSA and the weights-and-measures body; nominate an executive sponsor and a legal adviser.

2. Background and problem statement

2.1 The corridor today

The Northern Corridor carries port cargo from Mombasa to Nairobi and the region, and a large share of non-port traffic from mines, quarries, cement and sugar works, fuel depots and farms. Axle-load control is enforced at static weighbridges (Mariakani, Mtwapa, Athi River, Gilgil and others) under the EAC Vehicle Load Control Act and the 2018 Enforcement Regulations. High-speed weigh-in-motion (WIM) pre-screens traffic at some stations, but a static weighing is still the legal proof of overload, paper certificates and release orders travel with the driver, and every station repeats checks that an earlier station already made.

2.2 Where the chain duplicates checks and leaves gaps

Stage	Checked today	Duplication or gap	RCS v2 response
Before arrival	Manifest and ACD hold declared cargo weight	Declared weight is never compared with measured truck weight	Compute expected gross weight from declarations, container tare and truck tare, then compare with measurement
Port entry	Driver ID, licence, truck and trailer registration, access credentials, loading barcode	Same identity data re-checked at loading and exit	Enrol once; recognise by plate plus RFID tag
Loading	Container number matched to loading instruction; condition check; twist locks	Container number checked again after loading and at exit	Capture once at the gate-out portal with OCR and attach it to the token
Exit inspection	Container number, registrations, loading slip, seal, scanner, customs, shipping-line and KPA release, i-Gate validation, exit slip, KRA and KPA security verification	Releases are electronic, yet two agencies re-verify physically and by document	Show one release-status record to both agencies; physical checks only on exceptions
Port exit to road	No weight is measured	The first weighing happens far from cranes, customs and yard space	Weigh at the exit lane
Weighbridges	WIM, then static; paper certificate signed and stamped; presented again at later stations	Repeated at each station	Digital certificate and token lookup
Transit cargo	E-seal (ECTS), certificate at every station	An overloaded transit vehicle waits for a customs officer to break the seal	Registry alerts customs the moment an overload is confirmed
Overload handling	Fees, detention, paper release order carried through the journey	Later stations rely on the paper	Digital release order visible to every station

2.3 Problem statement

Compliant trucks lose time at every weighbridge because the system has no trusted record of an earlier weighing, overloaded trucks are discovered where correction is most disruptive, and high-speed WIM alone cannot supply

the dimension, axle-configuration, permit and certificate checks that KeNHA performs at static bridges. The result is queues, fuel burnt idling, repeated manual verification and weak data for enforcement and planning.

2.4 The evidence: what published data shows

Every figure below is published by an agency or the Northern Corridor Transit and Transport Coordination Authority (NCTTCA) and is held in the platform's data register with its period, source and classification.

Indicator	Value	Period	Source
Port of Mombasa throughput	45.45 Mt (+10.9 %); 2.11 M TEU, over 35 % empty	2025	NCTTCA 21st edition
Containerized import cargo dwell	106.7 h (target 48 h); 87 h in Jul-Sep 2026 (target 60 h)	2025; Q3 2026	NCTTCA
Customs release / after-release	57 h / 79 h	2025	NCTTCA 21st edition
Port truck turnaround	6.7 h (7.1 h in 2022)	2023	KPA annual bulletin
Mariakani trucks weighed per day	7,954-9,184 (2,501-2,720 a year earlier)	Jul-Sep 2026	KeNHA via NCTTCA
Mariakani compliance	99.1-99.9 %	Jul-Sep 2026	KeNHA via NCTTCA
National axle-load compliance	98.43 % over 10,776 km	FY 2024/25	KeNHA Annual Report 2025
Weighbridge crossing time	about 4 min corridor average; Mariakani 29 min during road works	2025	NCTTCA 21st edition (GPS)
Mombasa-Nairobi transit	66 h over 482 km (target 29 h)	2025	NCTTCA 21st edition (RECTS)
Mombasa-Malaba transit	94 h (target 55 h)	Jul-Sep 2026	NCTTCA (URA RECTS)
Agencies issuing permits	about 26	2024	Shippers Council of Eastern Africa

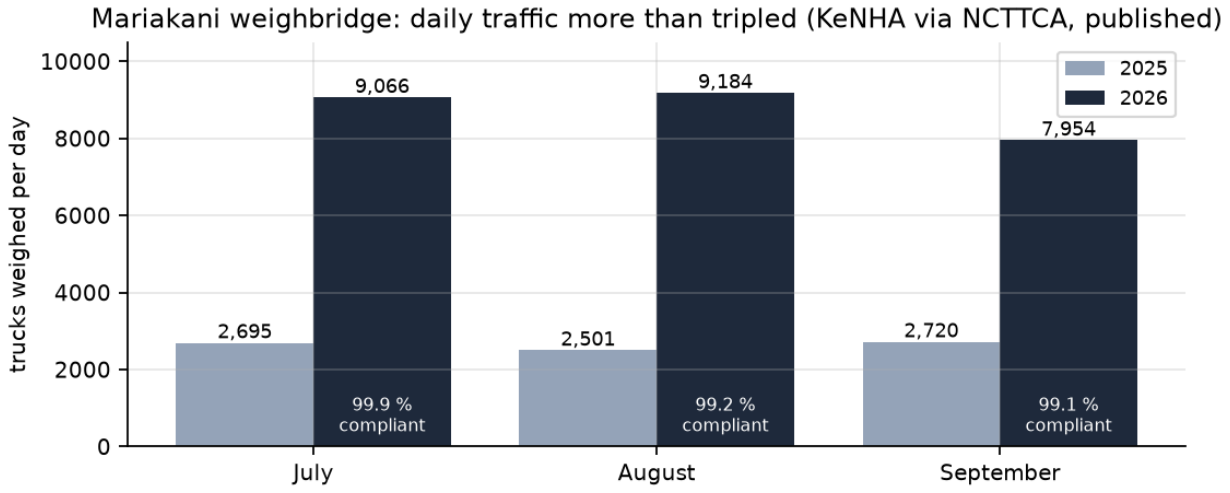


Figure A. Mariakani daily weighed traffic more than tripled year on year while compliance stayed near 99 % (published).

Reading it: almost every truck is compliant, yet every truck is weighed again at every station, and the port and corridor run well above their dwell and transit targets. The opportunity is to stop re-checking compliant trucks and to move checks to where they are cheapest.

3. Objectives

Main objective: remove repeat stops for compliant heavy trucks on the corridor by weighing each truck once at its first trusted point and honouring that result downstream, while keeping the static scale and the authorised officer as the legal reference for overloads.

Specific objectives:

- Design and build the port gate-out weigh node, origin weigh nodes, road screening nodes, the rolling verification lane and the clearance registry, with the Load Compliance Token as the single record that links them.
- Cover every parameter KeNHA checks at a static bridge (gross and axle-group loads, axle count and lifted axles, left-right balance, dimensions, permits, certificates, release orders, seals) through sensors and registry checks.
- Anchor every decision in the 2018 Regulations: screening never replaces static weighing as proof, officers e-sign red cases, calibration is quarterly, devices are approved.
- Quantify the effect with a validated four-level simulation (sensor, process, road traffic, system) and a go/no-go gate before hardware spend.
- Run a shadow pilot (port exit lane, one origin site, one verification lane at Mariakani) and only then a live pilot with volunteer fleets.

4. Proposed solution: RCS v2

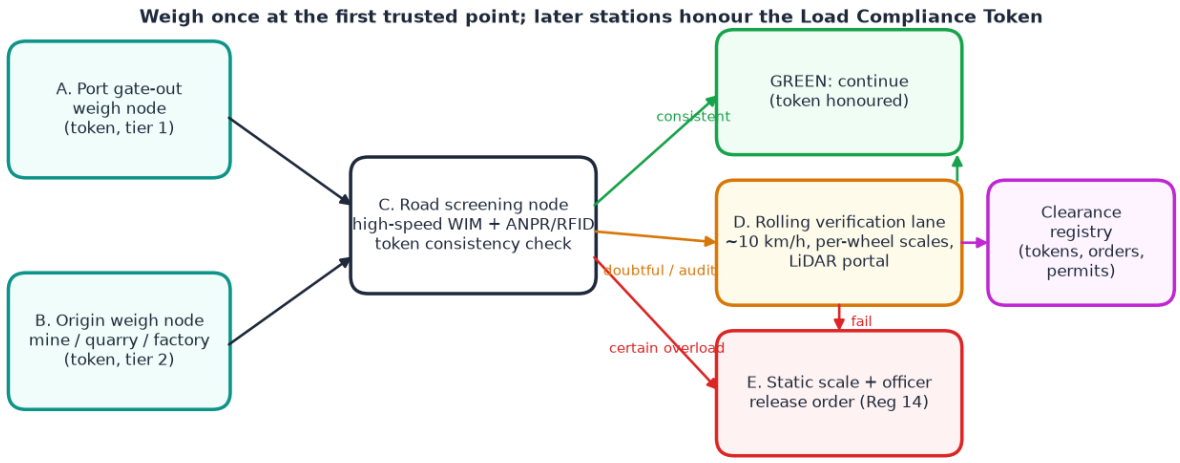


Figure 1. The RCS v2 clearance chain: five checkpoints and one registry.

4.1 Three truck streams, one rule

Stream	Examples	First trusted weight	Main risk
A. Port trucks	Containers from Mombasa	Port gate-out weigh node (tier 1)	Cargo changes after the port (low)
B. Industrial origin trucks	Mine, quarry, cement, sugar, fuel depot, farm and factory loads	Approved scale at the loading site linked to the registry (tier 2)	Under-reported weights at private scales; bulk loads that shift or settle
C. Untokened trucks	Spot loads, traders, small fleets, first-time users	First RCS road station or the rolling verification lane (tier 3)	No earlier weight to compare, so more uncertainty

A signed Load Compliance Token is issued at a truck's first verified weighing point and every later station honours it until it expires or a consistency check fails. Trust tiers set the audit rate: tier 1 (port) 2 %, tier 2 (approved origin scale) 4 % falling with a clean record, tier 3 (road node or lane) 3 %, tier 4 (on-board record only) always verified in the lane at first. Audit rates also rise with demerit points, so enforcement effort follows risk.

4.2 The Load Compliance Token

Fields: tractor and trailer plates and RFID tag; container number (check digit verified) and e-seal; axle-group loads, gross weight, expected weight and their variance; issuing node, time stamp and validity window; a snapshot of customs, shipping-line and KPA release status; route, destination and the declared driver; dimensions, axle configuration and permit reference; token source and trust tier; and an Ed25519 signature so

the record cannot be altered. Road nodes cache active tokens, roughly 8 000 a day.

4.3 Parameter coverage beyond weight

Parameter checked at a static bridge	High-speed WIM alone	Added in RCS v2
Gross vehicle weight; single axle and axle-group loads	Estimate only	Rolling verification lane at about 10 km/h on per-wheel scales (error target 1 %). Static scale remains legal proof for red cases
Axle count, configuration, lifted axles	A lifted axle looks like a missing one	Side LiDAR and camera classification matched to the registered configuration
Left-right imbalance	Only with per-wheel strips	Per-wheel scales in the lane, wheel-track position from WIM rows
Height, width, length	Not measured	Overhead LiDAR portal, light curtain for over-height
Special-load permit	Not checked	Plate links to a digital permit registry; scanner checks dimensions against the permit
Identity, licence, insurance, roadworthiness	Plate only	ANPR, RFID and trailer pairing; registry links to NTSA and insurers later (flagged, not blocking)
Earlier certificate, release order, transit seal	Not checked	Held digitally in the registry and shown to every station; e-seal status from KRA; customs alerted on confirmed overload
Fee, certificate, signature, calibration status	Not done	Registry computes the fee from the schedule and allowance policy; an officer e-signs; calibration register with quarterly certificates

4.4 The Rolling Verification Lane

A lane beside each station, reached by a diverge from the mainline screening node. Doubtful trucks and random audits roll through at about 10 km/h over low-speed per-wheel scales, under a LiDAR dimension portal and cameras; the full parameter set is captured in 20-30 seconds with no stop. At 2.8 m/s and about 27 m per truck (17 m truck plus a 10 m gap) one truck passes every 9.7 seconds, about 370 an hour, against roughly 15 an hour for a static lane. Layout assumption: 100-150 m approach with speed signs, a flat concrete measuring zone of at least 15 m and a 50 m exit, about 250 m in all; land and civil works must be confirmed early. If the lane's scales gain weights-and-measures approval, only red trucks need a static scale; until then, trucks measured within the margin of the limit still do.

4.5 Decision rules

Where	Result	Conditions
Port	Green	Identity and releases valid, gross and axle-group loads below the limit after the margin, measured and expected weight within tolerance. Token issued
Port	Amber	Near the limit or weight variance too large. Re-weigh in the port, correct if needed (customs present for sealed loads)
Port	Red	Over the limit beyond the margin. Held for corrective action (blocking gate-out is a policy decision for KPA, KRA and KeNHA)
Origin	Green	Approved scale reading below the limit after the margin, identity match, signed ticket. Origin token issued
Road mainline	Green	Valid token and identity match, weight within tolerance of the token weight (about 8 % for port or origin tokens, 11 % for road tokens), dimensions inside limits, axle count matches. Or no token but weight below the limit by the full margin
Road mainline	Divert to lane	No token and weight inside the margin band, inconsistent weight, dimension or axle-count mismatch, unreadable identity, or random audit
Verification lane	Green	Full parameter set within limits after the margin. Token issued, valid downstream
Verification lane	Red	Over the limit beyond the margin, dimension or permit breach, or inconclusive. Static scale and officer; hazardous and perishable flags change the handling

For loads in the margin band the road reading no longer has to be accurate against the limit; it only has to confirm that the weight is unchanged since the first trusted weighing, which is a much easier test.

4.6 What changes from the earlier proposal

Earlier	Now	Why
Weigh only on the road	Weigh at each truck's first trusted point and honour it downstream	Gives a reference and removes repeat stops
Port trucks only	All heavy trucks in three streams	Mines, quarries and factories are part of the traffic
Weight only	Weight, dimensions, axle configuration, lifted axles, per-wheel balance, permits, certificates	High-speed WIM misses most of what KeNHA checks at static bridges
Judge absolute weight against the limit	Compare with the token, absolute rule as fallback	Removes the near-limit accuracy problem for tokened trucks
Build our own road high-speed scales	Reuse KeNHA's high-speed WIM feed, add our own verification lane	Lower cost, and the lane supplies what the feed lacks
Static scale for every doubtful truck	Rolling lane for doubtful trucks, static scale for red cases	One rolling lane equals about 25 static lanes
Assumed pre-screening is acceptable	Use Reg 10(2), seek approval under Reg 5, approved signs under Reg 9	A legal basis for green decisions
Simulate one road station	Simulate the whole chain for three truck streams	The gain comes from moving and merging work

4.7 SmartPort: the eleven-stage port visit

RCS v2 decides what happens on the road; SmartPort organises everything before it. One port-visit record follows each truck and checks that each stage carries the data it needs before the next can start. Stage 7 is the port gate-out weigh node: when a loaded truck crosses it, the visit records gross and net weight, the discrepancy against the declared weight and the token, with no manual entry.

#	Stage	Key data captured	Owner
1	Pre-arrival registration and scheduling	Truck, driver, container, cargo type, declared weight, destination, appointment	Transporter, clearing agent
2	Customs clearance and document verification	Declaration, bill of lading, duty, inspection and permit status, hold reason	KRA, KenTrade, KEBS
3	Arrival and identification	ANPR plate, RFID tag, appointment validity, queue position and wait	KPA
4	Gate-in and security	Gate, access authorisation, vehicle inspection, driver verification, processing time	KPA
5	Yard allocation	Terminal, block, container position, loading instruction, yard wait	KPA
6	Cargo verification and loading	Equipment, seal, match against documents, condition, loading time	KPA
7	Automated weighing	Gross, tare, net, axle loads, discrepancy, calibration, Load Compliance Token	KPA, KeNHA
8	Final clearance	Release status, delivery order, gate pass, payment	KPA, KRA
9	Gate-out	Exit gate, cargo verified, exit authorised, port dwell time	KPA
10	Inland transport	Route, distance, ETA, stoppages, delay reason	Transporter, KeNHA
11	Delivery and trip closure	Arrival, proof of delivery, total transit time	Cargo owner

Agencies keep their own systems: KPA (port operations), KRA (customs), KenTrade (TradeNet single window), KeNHA (axle loads), NTSA (vehicle registration and roadworthiness) and KEBS (product standards). SmartPort complements them through authorised adapters and a shared dashboard, so each agency sees release, weighing and location status without re-keying documents.

5. Legal and regulatory basis

The EAC Vehicle Load Control (Enforcement Measures) Regulations, 2018 set the rules. The platform encodes them as configuration flagged for verification, never as assumptions buried in code.

Regulation	What it says	Design consequence
Reg 3(3)	The roads authority may outsource weighbridge operation to private operators	A legal route for a private partner role
Reg 3(4), 11(3)	Tickets and release orders for overloaded vehicles come from authorised officers	The platform prepares everything; the officer e-signs red cases (MFA)
Reg 4(5)-(6)	Mobile weighbridges and surveillance technology may be used; a vehicle in breach is escorted to the nearest static weighbridge	RCS readings decide who needs a static weighing; they never replace it as proof
Reg 5	Any weighbridge needs approval and a certificate from the weights-and-measures body	Port and lane scales need approval; a legal-grade flag gates what a lane reading may settle
Reg 6 and 8	Calibration at least every three months by an accredited calibrator; audit every twelve months	Quarterly calibration register; expired calibration blocks green
Reg 9(1)	Duty to submit for weighing arises on direction by an officer or a road sign	Driver signals must be approved signs
Reg 10(2)	A vehicle previously confirmed within limits by screening devices need not produce the paper certificate	The legal hook for the digital token replacing the paper certificate
Reg 10(4)	Certificates from any weighbridge are accepted as proof unless forged or tampered	Tokens and certificates are signed and tamper-evident
Reg 14	Release orders travel with the journey; a repeat overload in the same journey triples the fee	Digital release-order registry with repeat detection in the journey window
Reg 18	Transit certificates shown at every later weighbridge; overloaded transit vehicles detained until customs breaks the seal	Customs alerted automatically on a confirmed transit overload
Reg 20, Fifth Schedule	Demerit points against vehicle, transporter and driver	Demerits held in the registry and fed into audit rates
Second Schedule	Certificate records measured load, allowed load, a discretion allowance below 5 %, adjusted load, overload and fee	Allowance is a policy value to agree with KeNHA; it changes the stop share materially
Third Schedule, Reg 13	Legal weights (steering 8 t, single drive 10 t, tandem 18 t, tridem 24 t); fee table (5 000 kg over gross about 2 990 USD); detention free three days then 50 USD a day	Values in configuration, flagged for verification; strong incentive to join the programme

Two limits follow. A reading from screening devices cannot replace static weighing as proof of overload (Reg 4(6), 18(3)), and fees and release orders for overloaded trucks must come from an authorised officer. Open legal questions: whether Reg 10(2) covers a digital status, who must approve the devices, whether an officer may e-sign remotely, and whether low-speed in-motion weighing can be approved as legal-grade.

6. System architecture and current implementation

6.1 Architecture

Element	Where	Job
A. Port gate-out weigh node	KPA exit lane, beside the i-Gate	Low-speed weighing of every loaded truck; reads plate, RFID tag, container number and e-seal; compares measured with expected weight; issues the token
B. Origin weigh nodes	Mines, quarries, factories, depots	Approved scale at the loading site linked to the registry; issues an origin token
C. Road screening nodes	Mariakani, Athi River, Mtwapa and other stations	Existing high-speed WIM feed plus plate, tag and LiDAR readers; checks tokens and issues road tokens
D. Rolling verification lane	Beside each road station	Low-speed scales, dimension portal and cameras; captures every static-bridge parameter without stopping

Element	Where	Job
E. Static scales	Existing weighbridges and a port re-weigh lane	Legal proof for red cases and audits; results feed calibration
Clearance registry	Cloud, with edge caches at each site	One record per truck-journey: tokens, weights, dimensions, certificates, permits, release orders, demerit points, release status

6.2 What is already built

The project is not a blank sheet: the software platform exists, is tested and is deployed. The table lists the state on 05 November 2026.

Component	State	Notes
Clearance engine (clr-2.0.0)	Built, 78 API tests	Deterministic rules: token consistency, dimension, lifted axle, permit, discretion allowance, tiered audits, route (continue / lane / static / port re-weigh / hold); evidence hash per decision
Clearance registry API	Built	Load Compliance Tokens (issue, honour, suspend, public verify), Reg 14 release orders with officer e-signature and repeat tripling, special-load permits, trust-tier policy, stream and route statistics
Legal configuration	Built, flagged	2018 Regulations schedule, allowance, dimension limits, indicative fee table, detention terms; every value marked for verification
Web application	Built, deployed	Control centre, weighbridge console, clearance registry, RCS v2 model explorer, transporter, driver, cargo owner, warehouse, admin, public verify and About pages, 3D digital twin with sequential truck flow
Mobile apps	Written, not compiled	Flutter driver (hands-free) and cargo-owner apps
Roadside gateway	Built, 22 tests	Store-and-forward outbox, local clearance mirror, identity adapters, VMS driver
Firmware and hardware	Designed, not fabricated	STM32H743 + ADS131M08 acquisition firmware, ESP32 auxiliary node, analog front end, power, enclosure, BOM, verification-lane and port-node layouts
Simulation package	Built, 32 tests	RCS v2 chain Monte Carlo, experiments (replications, sweeps, tornado, gate), SimPy station model with tokens and lane, SUMO adapter
Cloud	Deployed	Vercel (functions in Dublin) + Supabase PostgreSQL (Ireland); AWS production design in Terraform, validated, not applied
Security	Built	RBAC, MFA for privileged roles, Argon2 passwords, Ed25519-signed tickets/tokens/orders, hash-chained ledger, PII encryption, DPIA under the Data Protection Act 2019

Honesty rules carried by the platform: every weighing record carries a source of live, simulated or replay and simulated data is never shown as live; government adapters are mock until credentials are authorised; environmental and chain-model figures are labelled modelled; accuracy and latency targets remain targets until measured.

6.3 Data provenance

Every parameter carries one of four classes: actual (operational records), published (agency or NCTTCA reports), estimated (design assumptions) or simulated (model output). The register currently holds 52 entries; the port gate service times and gate-side truck volumes are estimated because they are not published, and are first in line for replacement with KPA gate logs. The web platform shows the class beside every figure, and simulated records are never displayed as live.

7. Simulation methodology

7.1 Questions the simulation answers

- How many stops and how much delay does each design remove?
- How accurate must the port and road scales be?

- How robust is the system to low token coverage, outages and traffic peaks?
- Can the port re-weigh stage cope with the volume?
- Do queues at the port exit and the weighbridge stay stable?

7.2 Four linked levels

Level	Simulates	Tools	Output
L1 Sensor and algorithm	Truck dynamics on a rough road, tyre contact, sensor response, temperature drift and noise, then the weight estimator	Python (NumPy, SciPy), quarter-car model	Error curves by speed, load and temperature; they set the port and road errors used in L2
L2 Process	The whole chain for three truck streams, as-is and to-be, as a discrete-event model	SimPy; the Monte Carlo chain model	Stops, delays, queues, truck-hours, false-clear rate
L3 Road traffic	Approach lanes, diverge lanes, the verification lane and queue spillback onto the mainline	SUMO with TraCI	Queue lengths, spillback risk, diverge lane length
L4 System and hardware-in-the-loop	Token service, APIs and edge pipeline under load and failure	FastAPI mock agencies, Locust, Mosquitto MQTT, recorded signals replayed into the bench rig	Latency, failover behaviour, data integrity

7.3 The two L2 models used in this proposal

The chain model is a vectorised Monte Carlo of the whole clearance chain. Each truck draws a true load from a normal distribution with mean equal to the average loading (75-90 % of the legal limit) and a spread of 12 % of the limit, capped just below the limit so that heavily loaded fleets bunch at it; 1.5 % of trucks are overloaded (from the reported 98.5 % compliance), uniformly 0-15 % over. Every node reads the true load with its own relative error: port 1 %, origin 1 %, road 2.5 %, lane 1 %. A screening node sends a truck for a closer look when reading x (1 + 2 x error) exceeds the limit (expanded uncertainty, k = 2, the same convention as the clearance engine). A tokened truck on the road is only checked for consistency: it passes when the road reading is within three combined standard deviations of the token weight, about 8 % for port and origin tokens and 11 % for road tokens. Random audits of 3 % go through the lane. Common random numbers make the comparison between policies paired rather than noisy.

The station model is a discrete-event queue model in SimPy: non-homogeneous Poisson arrivals with an hour-of-day profile, two high-speed WIM lanes, five static bays with a four-minute median service, two inspection bays, and in the RCS v2 policy a rolling verification lane resource (one lane, 90-second median pass), token coverage, token error, a share of trucks whose cargo changed after the token, and lane closures. Decisions come from the production clearance engine so the simulation and the live platform can never disagree on policy. The station model keeps a heavier synthetic population (15 % overloaded) than the chain model, so its absolute shares differ; it is used for queues, lane capacity and closure behaviour, not for the headline stop shares.

7.4 Experiment design, verification and validation

- Factors: average loading 75-90 %, port error 0.5-2 %, road error 1.5-4 %, token coverage 50-100 %, peak factor 1.0-2.0, outage frequency and duration, audit share 1-5 %.
- 30 replications per scenario with different seeds and common random numbers; means reported with 95 % confidence intervals (Student t).
- KPIs: stops per 1 000 trucks, mean and 95th-percentile delay, truck-hours a day, maximum queue, false-clear rate, port re-weigh load, port yard dwell effect.
- Verification: unit tests, conservation checks (every truck exits once), hand-traced single trucks, comparison of a single-server version with the Erlang C formula.

- Validation: compare simulated as-is delays and queues with measured baselines; accept if the mean delay is within about 15 % and peak queues align; review the model with KPA and KeNHA staff. A tornado chart shows the committee which assumptions matter most.
- If data access is delayed: a two-week time-and-motion and counting study with cameras and stopwatches, results flagged provisional.

8. Simulation results (modelled)

All figures in this section were generated from the repository's simulation package on 05 November 2026 with the assumptions stated in section 7. They are model outputs, not measurements, and they are the author's estimates to be replaced by observed data in Phase 0-2.

8.1 Port trucks: effect by average loading

Average load (% of limit)	Road-only screening	Road-only with 5 % allowance	RCS v2 road stops	RCS v2 port re-weigh	Road truck-hours saved per day (v1 to v2)
75 %	9.0 %	5.7 %	3.8 %	4.2 %	about 41
80 %	14.4 %	7.5 %	4.4 %	7.9 %	about 80
85 %	23.2 %	10.8 %	5.3 %	14.8 %	about 143
90 %	35.6 %	16.1 %	6.5 %	25.6 %	about 232

Truck-hours use 8 000 trucks a day and 6 minutes per stop, so one percentage point is 8 truck-hours. 600 000 trucks per row.

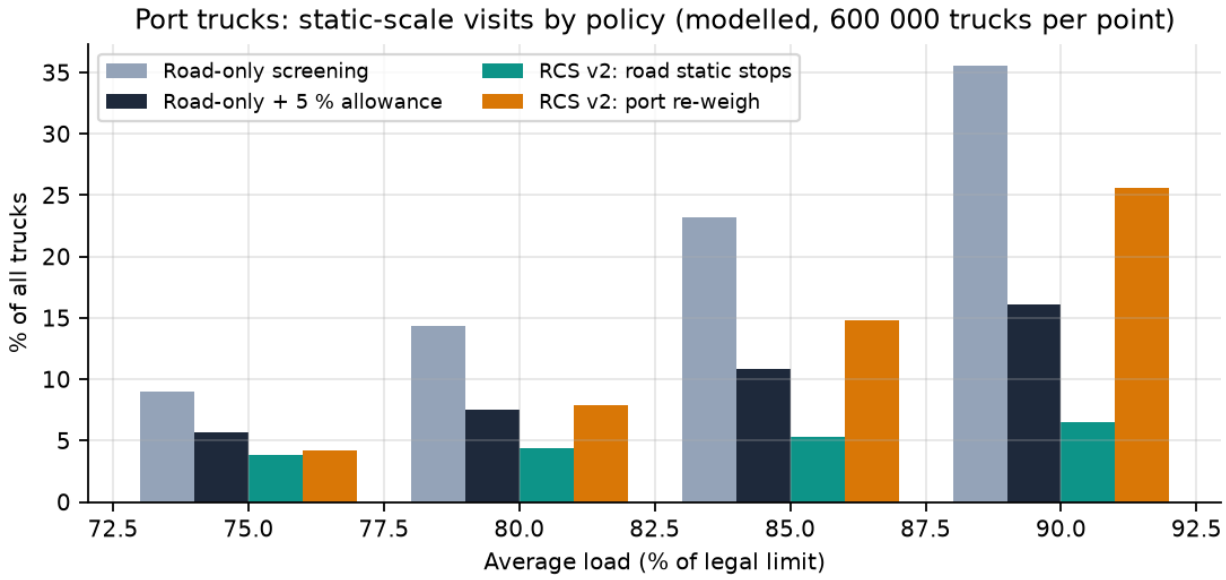


Figure 2. Static-scale visits of port trucks by policy and average loading (modelled).

Reading it honestly: v2 does not delete the near-limit checks, it moves them from the public road to the port, where trucks are already in a controlled area with cranes and customs. At 85 % loading there are still about as many re-weighs in total (5.3 % on the road plus 14.8 % at the port versus 23.2 % on the road). The saving comes from relieving road queues and from fixing overloads where it is cheap, and the more heavily loaded the fleet, the larger the shift.

8.2 Mixed traffic: effect by token coverage

Trucks with a port or origin token	First station: sent to the lane	First station: static scale	Road-only baseline (static)	Later stations: lane	Later stations: static	Truck-hours saved per day at the first station
30 %	17.2 %	10.0 %	20.8 %	3.3 %	0.01 %	about 52
50 %	13.2 %	7.1 %	20.8 %	3.3 %	0.02 %	about 83
70 %	9.2 %	4.3 %	20.8 %	3.3 %	0.03 %	about 114
90 %	5.3 %	1.5 %	20.8 %	3.3 %	0.04 %	about 144

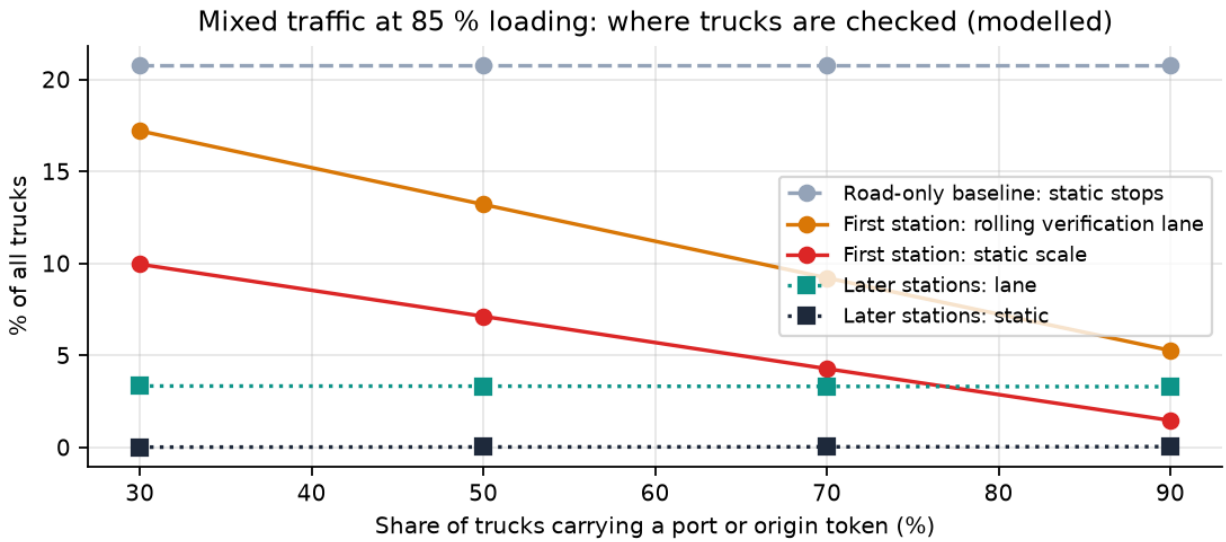


Figure 3. Where trucks are checked at the first and at later stations as token coverage rises (85 % loading, modelled).

- Later stations: once every truck carries a token (port, origin or road), about 3.3 % go to the lane (audits and mismatches) and almost none to a static scale. This is where the repeat stops disappear.
- Untokened trucks alone: the lane trims static-scale visits only from about 21 % to 14 % of those trucks. The rest are loaded within the 2 % margin of the limit and need either a legal-grade lane reading or the discretion allowance.
- Capacity: at 500 trucks an hour the modelled lane peak is about 116 trucks an hour against a capacity near 370, so one lane per station is enough.
- Fleet enrolment is therefore a core workstream: token coverage is the lever that moves the road stop share most.

8.3 Sensitivity

- Token coverage: at 85 % loading the road stop share is 13.2 % at 50 %, 8.2 % at 75 %, 5.3 % at 90 %, 3.3 % at 100 % coverage.
- Port scale accuracy mainly moves the port re-weigh share (13.0 % at 0.5 % error, 14.8 % at 1.0 % error, 18.7 % at 2.0 % error) and barely changes the road stop share.
- Discretion allowance at the port: re-weighs fall to 14.8 % at 0.0 %, 4.6 % at 2.5 %, 1.2 % at 5.0 %. Applying the allowance is a policy decision for KeNHA; approving the port scale as legal-grade has a similar effect.

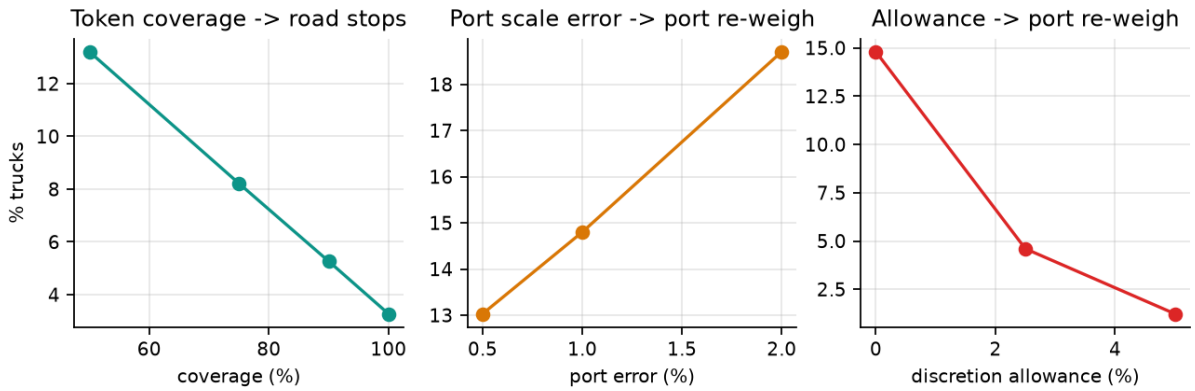


Figure 4. One-factor sweeps: coverage drives road stops; port error and allowance drive port re-weighs (modelled).

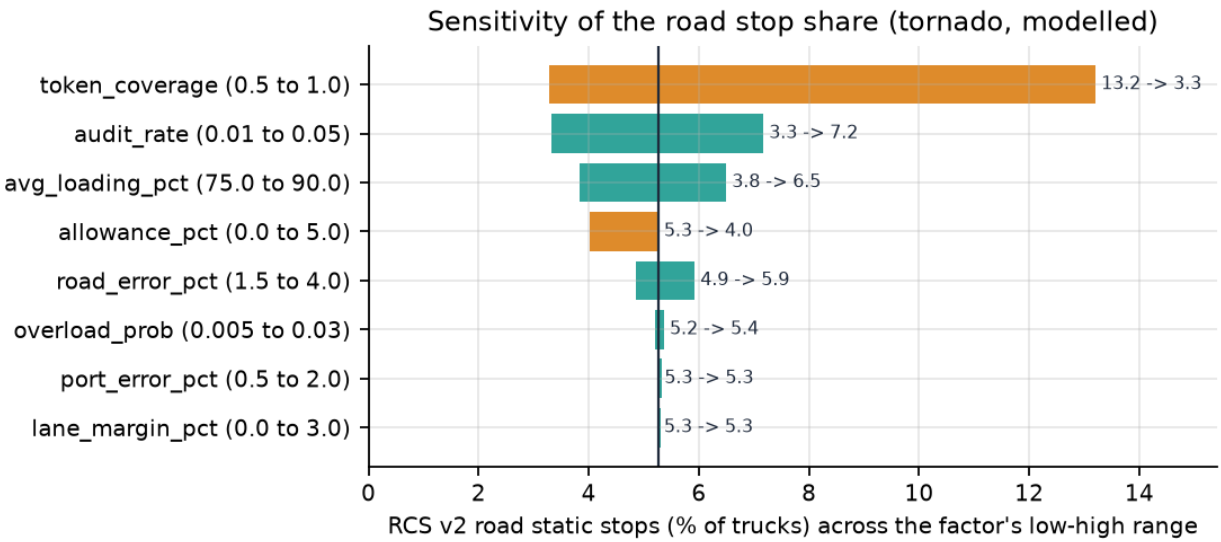


Figure 5. Tornado chart: swing of the RCS v2 road stop share across each factor's low-high range (modelled).

8.4 Replication and confidence intervals

Output (defaults: 85 % loading, 90 % coverage)	Mean	95 % confidence interval
road only static	23.32	23.28 - 23.36
road only with allowance	10.91	10.88 - 10.94
v2 road static	5.27	5.25 - 5.29
v2 port reweigh	14.92	14.89 - 14.95
first station lane	5.27	5.25 - 5.29
first station static	1.47	1.46 - 1.48
later station lane	3.27	3.26 - 3.29
later station static	0.04	0.04 - 0.04
untokened lane	23.23	23.11 - 23.35
untokened static	14.19	14.10 - 14.28
truck hours saved per day	144.41	144.15 - 144.68
truck hours saved first station	145.18	144.90 - 145.46
false clear rate	0.0011 %	0.0009 - 0.0014 %

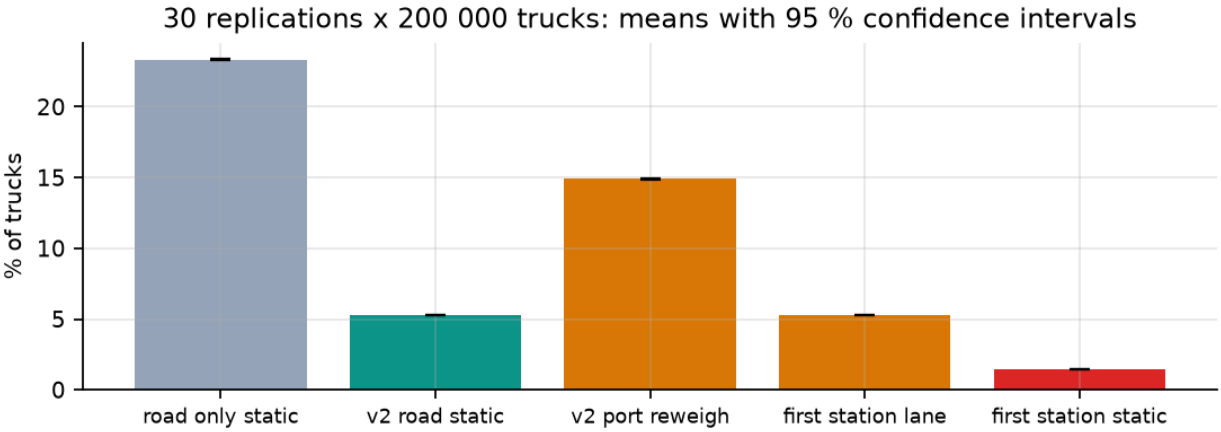


Figure 6. Means and 95 % confidence intervals over 30 replications (200 000 trucks each). Intervals are narrow: the uncertainty that matters is in the assumptions, not the sampling.

8.5 Go/no-go gate

Check	Value	Threshold	Result	Detail
road stop share cut points	17.90	10.00	pass	road-only 23.2 % -> v2 5.3 %
false clear rate	0.00	0.01	pass	share of trucks allowed through without a stop that were over the limit
port reweigh share pct	14.80	20.00	pass	port re-weigh volume must be practical for KPA (threshold to be agreed)
lane capacity	116.40	370.40	pass	peak lane demand against one rolling verification lane

The model meets the gate (proceed = yes). The gate is applied again to the validated model after the baseline study: proceed to the shadow pilot only if the validated model cuts the road stop share by at least 10 points against the measured baseline, or removes recurring peak saturation, with a false-clear rate at or below 1 % and a port re-weigh volume that KPA confirms is practical. Otherwise the project narrows to a vehicle-side or registry-only offering.

8.6 Station queue model (SimPy)

Scenario	Trucks (12 h)	Mean time in system (min)	P90 (min)	Max static queue	% stopped	% via lane	Static utilisation
baseline_all_stop	918	44.3	70.7	94	100.0 %	0.0 %	99 %
rolling_clearance	918	2.4	4.7	1	22.8 %	0.0 %	23 %
arrivals_+50	1400	3.3	4.8	3	21.6 %	0.0 %	33 %
rcs_v2_untokened	918	1.7	2.8	0	6.7 %	18.6 %	7 %
rcs_v2_tokens_50	918	2.0	3.0	0	7.1 %	19.8 %	8 %
rcs_v2_tokens_90	918	2.3	3.1	0	7.3 %	19.9 %	8 %
rcs_v2_allowance_5pct	918	1.9	2.6	0	6.1 %	20.0 %	6 %
rcs_v2_lane_closure	918	2.3	3.1	0	8.6 %	15.9 %	9 %

Means over 3 replications of 12 h; 60 trucks an hour mean demand with a synthetic hour-of-day profile. The all-stop baseline saturates the static scale (every truck stops); rolling clearance stops about a fifth; RCS v2 moves most of those into the lane without stopping and keeps the static scale below 10 % utilisation. The +50 % demand scenario shows rolling clearance alone starting to load the static scale again, which the lane absorbs. A two-hour lane closure sends doubtful trucks back to the static scale, as designed.

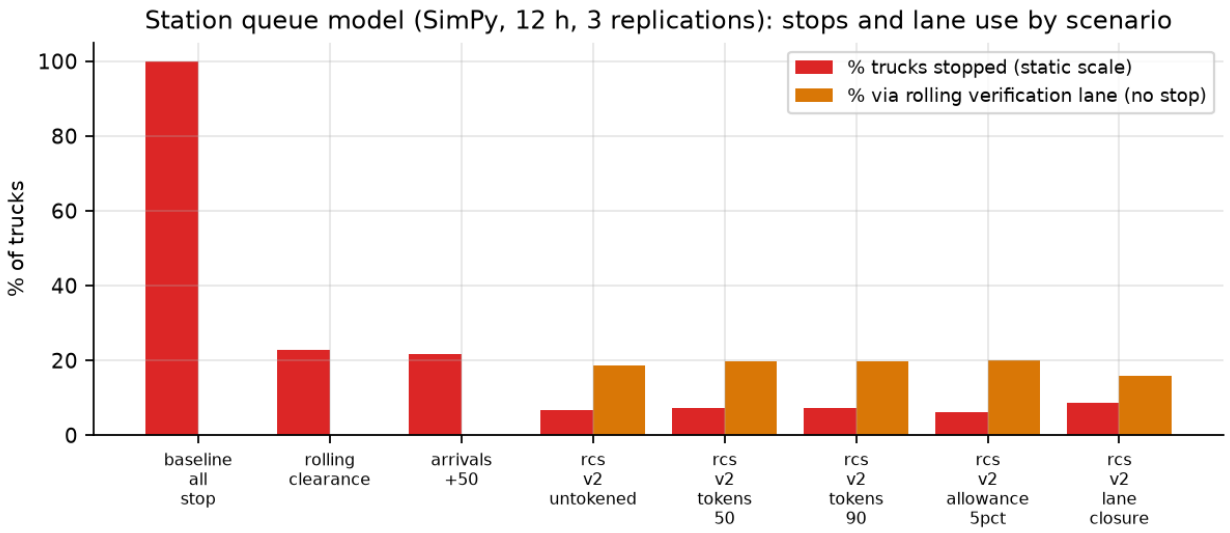


Figure 7. Share of trucks stopped at the static scale and share rolling through the verification lane, by scenario (modelled).

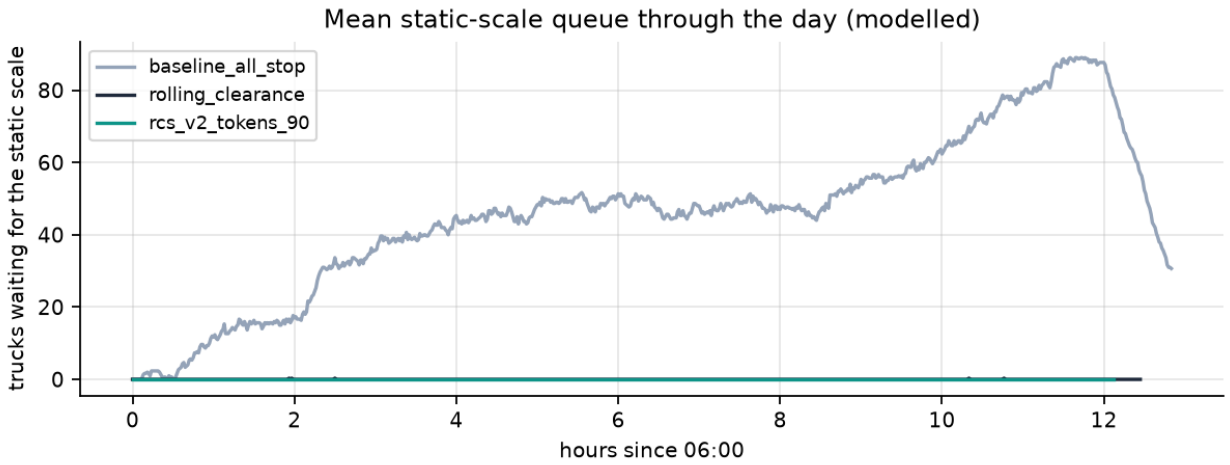


Figure 8. Mean static-scale queue through the day: all-stop baseline, rolling clearance, RCS v2 with 90 % token coverage (modelled).

8.7 Port process: existing operations vs SmartPort

A discrete-event model of stages 3-11 (gates, yard, loading, weighing, clearance with document holds, gate-out, inland transit), 5 replications. Existing-operations dwell 6.8 h against KPA's published 2023 truck turnaround of 6.7 h; the yard-wait median (315 min) is the tuned parameter.

KPI	Existing	SmartPort	Change
Mean port dwell (h)	6.81	3.97	-42 %
P90 port dwell (h)	10.80	6.33	-41 %
Queue waiting per truck (min)	1.88	0.00	-100 %
Weighbridge service (min)	5.14	1.55	-70 %
Gate-out throughput (trucks/h)	79.21	84.42	+7 %
Gate-in utilisation	0.60	0.15	-75 %
Weighbridge utilisation	0.62	0.19	-70 %

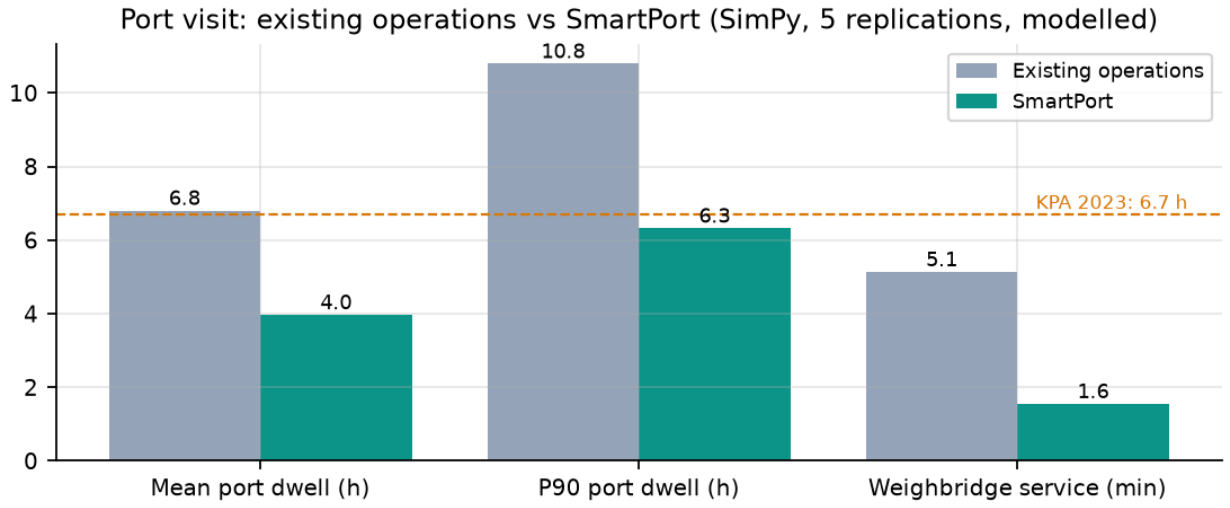


Figure 9. Port visit KPIs with the KPA 2023 truck-turnaround benchmark (modelled).

- 2400 trucks a day at the port gates (estimated; replace with KPA gate logs); day-time peak factor 1.5 unless appointments smooth arrivals.
- Gate service 240 s manual vs 60 s RFID/ANPR; weighing 300 s static vs 90 s at the port gate-out node.
- Document holds 15% existing vs 5% with pre-cleared data sharing; median hold 45 min; SmartPort clearance desk time halved.
- Yard wait median 315 min existing vs 180 min with readiness-based appointments (design assumption).
- Inland transit at 45 km/h over 482 km is a free-flow estimate; NCTTCA measured Mombasa-Nairobi at 66 h (RECTS, 2025), so most observed transit time is stoppage and seal-handling delay outside this model.

8.8 Not yet counted

Time saved from removed duplicate checks at port entry, loading and exit; fees and detention avoided by transporters; customs benefits from catching weight misdeclaration; fuel and emissions from avoided idling (the platform's environmental model labels these modelled). The baseline study must time each duplicate check before these are claimed.

9. Implementation roadmap

Phase	When	What happens	Exit gate
0-1. Access and legal	Months 0-3	Meet KPA, KRA, KeNHA, NTSA, KenTrade, the weights-and-measures body, the Kenya Transporters Association and two or three candidate loaders. Request data. Get a legal opinion on Reg 10(2), Reg 5, Reg 3(4) and digital certificates	MoUs or written interest, data access agreed, legal route clear
2. Baseline and design	Months 3-7	Time-and-motion study at the port exit and Mariakani; classification count for the real share of port and non-port trucks; study of dimension and lifted-axle violations; run and validate the simulation; specify token, APIs and security	Simulation shows the gate is met
3. Bench prototype	Months 6-11	Scale-model rig, token service, OCR on recorded video, LiDAR dimension portal test, per-wheel scale test, mock agency APIs, replay of recorded signals	Latency and repeatability targets met on the rig
4. Shadow pilot	Months 10-16	One port exit lane, one origin site and a verification lane at Mariakani; tokens issued and compared, nobody acts on them	Accuracy and false-clear targets met over a sustained period
5. Live pilot	Months 16-24	Volunteer fleets and loaders use the green lane; approved signs; static scale audits continue; seek approval of lane scales as legal-grade	Agency approval to operate live, then scale to other stations

9.1 Simulation work plan (16 weeks, Phase 2)

Weeks	Work
1-2	Baseline study plan and data requests
3-4	L1 sensor and algorithm model
5-8	L2 process model, verification and first scenarios
9-10	L3 road traffic model
11-14	L4 token service, mock APIs and hardware-in-the-loop
15-16	Validation, sensitivity, committee report

9.2 Integration map

System	Data exchanged	Dependency risk
KPA gate systems (i-Gate, KWATOS)	Gate events, release status, exit slip	KPA must allow a weigh node in the exit lane
KRA (iCMS, ACD, tracking seals)	Release status, declared weight, seal ID, overload alerts	Data-sharing agreement
KeNHA control centre and WIM feed	Road weights, static results, certificates, release orders, officer e-signature	Access to the feed or permission to add readers and a lane
Loading-site scales (private)	Signed tickets, calibration certificates	Approval and audit of each scale
NTSA vehicle and permit records	Registration, tare, axle configuration, special-load permits	Data quality
KenTrade and clearing agents; insurers (later)	Document status; insurance validity	Optional first, deeper later

9.3 Deliverables by phase

- Phase 0-1: stakeholder MoUs, data-sharing agreements, legal opinion, data-protection registration plan.
- Phase 2: baseline study report, validated simulation with sensitivity analysis, token and API specification, security design, go/no-go recommendation.
- Phase 3: bench rig results (latency, repeatability, OCR accuracy, dimension portal accuracy), hardware-in-the-loop report.

- Phase 4: shadow pilot accuracy report (WIM and lane versus static), false-clear and identity rates, token consistency statistics, origin drift detection record.
- Phase 5: live pilot KPIs, regulator pack, approval applications for the port and lane scales.

9.4 Cost drivers

A costed budget is prepared with stakeholders in Phase 0-1 once site access, the WIM feed arrangement and the number of lanes are known. The main drivers are: the port gate-out node (low-speed scale, readers, portal, civil works in the KPA exit lane), one rolling verification lane per station (about 250 m of approach, measuring zone and exit, per-wheel scales, LiDAR portal, cameras, cabinet and power), origin node readers and registry integration at each approved loader, quarterly accredited calibration and the twelve-month audit (Reg 6 and 8), weights-and-measures approvals, cloud hosting and support, the baseline study, and fleet enrolment (tags, onboarding). Reusing KeNHA's high-speed WIM feed rather than building road scales is the largest avoided cost.

10. Risks and mitigations

Risk	Mitigation
KPA may not allow a weigh node in the exit lane, or may not accept weight as a gate-out condition	Early engagement; the port node can start as advisory (shadow); the design also works with the first road station as the first trusted point
A port or lane scale needs weights-and-measures approval; without legal-grade status, near-limit trucks still need static weighing	Seek approval early; model both cases; apply the discretion allowance where KeNHA agrees
Private origin scales may under-report	Approval and audit of each scale; comparison of origin readings with road and lane readings; token suspension for drift
Token fraud: swapped plates or cargo added after the first weighing	Bind the token to plate, tag and seal; weight consistency on the road; audits; demerits
Legitimate weight changes (partial offloads, fuel use)	Validity window and tolerance; a lane pass renews a token
Untokened and first-time trucks carry more uncertainty	Verification lane; fleet and loader enrolment as a core workstream
Land and civil works for the lane at each station	Confirm availability early; enforce speed discipline with approved signs
Officers must still sign tickets and release orders (Reg 3(4), 11(3))	E-signature workflow with MFA; legal opinion on remote signing
Data-sharing delays and cybersecurity	Signed records, audit logs, DPIA under the Data Protection Act 2019, two-week provisional study if data is late
Model risk: assumptions differ from reality	Validation against measured baselines before any claim; tornado analysis; gate re-applied on validated model

11. Monitoring and evaluation

Indicator	Target (engineering target until measured)	How measured
False-clear rate among green decisions	at most 1 %	Shadow pilot: green decisions paired with static results
WIM and lane accuracy	95 % of gross errors within +/-5 % (WIM); lane error target 1 %	Paired passes per lane, speed band and temperature band
Identity match rate	at least 97 %	Confidence at or above 0.90 and correct vehicle versus operator-confirmed plate
Decision latency at the edge	at most 2 000 ms at P99	Gateway latency distribution
Road stop share at the first station	at least 10 points below the measured baseline	Station logs before and after
Token consistency rate	reported; suspensions investigated	Registry statistics
Origin scale drift	detected and acted on	Origin readings versus road and lane readings

Indicator	Target (engineering target until measured)	How measured
Availability	99.9 % for the registry and nodes	Monitoring

12. Decisions requested

1. Approve the redesign around the port gate-out weigh node and the Load Compliance Token.
2. Approve Phases 0-2, including the 16-week simulation and the baseline study, with a gate before any hardware spend.
3. Authorise approaches to KPA, KRA, KeNHA, NTSA and the weights-and-measures body.
4. Nominate an executive sponsor and a legal adviser.

Appendix A. Data requests

Dataset	Owner	Use
Gate-out and i-Gate timestamps, exit slips	KPA	Port exit delay and duplicate checks
Weighbridge logs: time, gross and axle loads, outcome	KeNHA	Loading distribution, baseline stops and delays
Release timestamps, declared weights sample, tracking-seal events	KRA	Expected weight, misdeclaration rate, transit flows
Tare weights and axle configurations	NTSA	Expected-weight calculation
Fleet and trip-plan information	Kenya Transporters Association	Enrolment and coverage estimates

Plates and driver details are anonymised before analysis, in line with the Data Protection Act, 2019.

Appendix B. Open questions to confirm

- What share of trucks at Mariakani and other stations come from the port, from industrial origins and from elsewhere?
- Does KeNHA already share road WIM results with KPA or KRA, and can the raw feed be obtained?
- Does the port already weigh imports at the gate?
- Which dimension limits, lifted-axle and permit checks does KeNHA enforce at static bridges today, and how often are they violated?
- Can the Reg 10(2) screening exemption apply to a digital token, who must approve the devices, and can an authorised officer e-sign remotely?
- Can low-speed in-motion weighing be approved as legal-grade? Is the 5 % discretion allowance applied routinely?
- Which route-specific limits apply, and does the 34-tonne container limit still hold?

Appendix C. Model assumptions

Parameter	Value	How to measure
Static scale lanes and service time	8 lanes (plan) / 5 bays (station model), 4 minutes	Observe at Mariakani, request logs
Extra time to stop and rejoin traffic	2 minutes	Stopwatch and cameras
Port exit validation time; duplicate checks	Unknown	Time-and-motion at the i-Gate; count and time each
Port scale approach speed; lane speed	10 km/h	Design choice
Token coverage; audit share	90 %; 3 %	Enrolment plan; policy
Share of loads near the limit	Mean 75-90 % of limit, spread 12 %	Weighbridge logs
Scale errors: port, origin, road, lane	1 %, 1 %, 2.5 %, 1 %	L1 model, calibration records, shadow pilot
Lane pass cost; static stop cost; headway	1.5 min; 6 min; 27 m at 10 km/h	Design assumption, to time in the pilot
Arrivals	8 000 trucks a day; peak factor 1.5; bursts after vessel discharge	KPA and KeNHA logs

Appendix D. Sources

- NCTTCA, 21st Edition Northern Corridor Transport Observatory Report, April 2026 (2025 data): top.ttcanc.org/uploads/attachments/cmownhhl201qpfmqhl5ikzprd-en-21-top-report.pdf

- NCTTCA, Quarterly Performance Report July-September 2026: top.ttcanc.org/uploads/attachments/cmvo7m7gv00qfwznu9zlgz9zh-quarterly-report-july-to-september-2026.pdf
- KeNHA Annual Report 2025: kenha.co.ke/wp-content/uploads/2026/06/KeNHA-ANNUAL-REPORT-2025-WEB-VERSION.pdf
- KPA annual bulletins via Kenya News Agency and The Star (2023 truck turnaround 6.7 h; 2024 throughput and ship times).
- Shippers Council of Eastern Africa (dwell history, permit agencies); KenTrade single window reports (agencies connected, document reduction).
- Full list with URLs and periods: docs/06-operations/data-register.md and the live Data register page.
- EAC Vehicle Load Control (Enforcement Measures) Regulations, 2018 (full text, bdears.busitema.ac.ug).
- Process guide supplied by the project owner covering ship arrival, customs, port release, truck exit and weighbridges, citing KRA, KPA, KeNHA, InfoTrade Kenya and the Kenya Transporters Association; reported gross limits and 2026 events not independently re-verified.
- Earlier RCS decision brief and hardware analysis: sensor, calibration and benefit assumptions.
- Repository documents: docs/01-requirements/rcs-v2-port-anchored-clearance.md, legal-hooks-eac-vlc-2018.md, docs/06-operations/rcs-v2-simulation-plan.md, simulation-guide.md, the engineering report and the requirements traceability matrix (https://github.com/Jrjunior-mk/KIFARU-PORTOS-1.0).
- Simulation outputs used here: docs/07-proposal/assets/chain-results.json and simpy-comparison.json, regenerated with make_charts.py and build_proposal.py.